

e-clearing.net
OCPI Connection Information

Anas Munir, Zubair Aslam

e-clearing.net
necessary conventions for OCPI

- **General Rules:**
  - Some attributes may be dropped if they cannot be mapped to OCHP
  - Data not compliant to the following rules, will be dropped entirely
- **Tokens:**
  - **M:** auth_id must contain a valid eMI3 Contract ID (**NLEVNC**1234ABCD)*
  - auth_id: Regular expressions: [A-Z]{2}-[A-Za-z0-9]{3}-[Cc][A-Za-z0-9]{8}-[A-Za-z0-9], [A-Z]{2}[A-Za-z0-9]{3}[Cc][A-Za-z0-9]{8}[A-Za-z0-9], [A-Z]{2}-[A-Za-z0-9]{3}-[Cc][A-Za-z0-9]{8}, [A-Z]{2}[A-Za-z0-9]{3}[Cc][A-Za-z0-9]{8}
  - auth_ids will be normalized (All upper case, no separators), check digit will be removed. Since eMI3 standard defines: „The eMA ID MUST be interpreted case insensitive, separators are optional and Check digit is also optional."
  - Empty strings or lists in required fields, or object just filled with spaces or not visible characters will be rejected.
  - If you want to put data as a sub-provider (you are DEABC for example and want to upload for BEABC), then you need to be in the same e-clearing frontend group.

M: = mandatory

* http://www.ochp.eu/wp-content/uploads/2014/07/Poster-eMobility-IDs.pdf

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- **Tokens:**
  - **M:** e-clearing only accepts whitelist: ALLOWED or ALWAYS, and type: RFID. If optional seperator i.e. „-“ is used in auth_id at one place, then it MUST be set at all three places.
  - Auth_id cannot be updated.
  - Only UTF-8 is accepted (No control characters).
  - Real time authorisation is supported but the check is done against e-clearing database and the messages will not be forwarded to the EMPs.
  - “uid” will be transformed to UPPERCASE. Means, CPOs will get the transformed “uid” in the tokens and EMPs will also get the transformed version back when they do a GET on their own Tokens.

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- **Locations:**
  - A new location must have at least one evse.
  - Capabilities of EVSE should be provided.
  - last_updated in Location is enforced from inside out. i.e. Location cannot be newer than EVSE/Connector.
  - EVSE ID/UID cannot be modified once created, EVSE ID/UID combination is unique and will not change.
  - EVSE ID/UID combination can be moved from one location to another, but the combination itself cannot be changed.
  - **M:** OCPI evse_id has to contain a valid eMI3 EVSE-ID (**NLEVNE**123456ABCD)*
  - **M:** OCPI tariff_id has to contain a valid OCHP tariffId (**NLEVNT**12345ABC)
  - evse_id is normalized in following manner:
    - Optional separators are removed from the id excluding from the Power Outlet ID. Optional Separators mentioned in Power Outlet ID are considered to be part of the id and hence aren‘t optional anymore. E.g. FR*A23*E45B*78C. Here Power Outlet ID is: “45B*78C”. Also: „FR*A23*E45B*78C“ & „FR*A23*E45B78C“ are consequently two unique evse_ids in e-clearing platform. Also it is transformed to UPPERCASE since as per eMI3: Alpha characters SHALL be interpreted case insensitively.
  - evse_id Regular Expression: [A-Z]{2}*[A-Za-z0-9]{3}*[E][A-Za-z0-9][A-Za-z0-9*]{0,30} // [A-Z]{2}[A-Za-z0-9]{3}[E][A-Za-z0-9][A-Za-z0-9*]{0,30}

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- **Locations:**
  - Postal code must be in uppercase ascii-7 characters.
  - Longitude/Latitude has to have a fixed length of 6 decimals.
  - UTF-8 without control-character is allowed.
  - Empty strings or lists in required fields, or objects just filled with spaces or not visible characters will be rejected.
  - If a list contains duplicates of anything the request will be rejected.
  - Patch requests on list elements are not well defined if there is no id available. If a list is patched it will get overridden completely and therefore we need all required data to do so.

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- **CDRs:**
  - **M:** CDR.id must contain the CPO-ID (**NLEVN**12345ABCDEFG…) [up to 36 characters total]
  - **M:** CDR.id must follow the pattern: Regular Expression: [0-9A-Z]{1,36}
  - The Location object must be provided with CDR as defined by the protocol.
  - OCHP data may not be sufficient to supply all mandatory attributes with meaningful information (please keep in mind that some systems will reject a CDR with value 0)

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- **Tariffs:**
  - **M**: Tariff_id must contain CPO id e.g. DELND.
  - **M**: Tariff_id must contain „T“ after CPO id. e.g. NLEVNTA1….. [up to 36 characters total]
  - Tariff_id: Regular Expression: [A-Z]{2}[A-Za-z0-9]{3}[T][A-Z0-9][A-Z0-9*]{0,29}
  - Only uppercase characters are supported in the tariff_id.
  - Not yet live, currently in the testing phase.
  - If all the restrictions in the Tariff are invalid/expired at the time of creation/upload, it will not be accepted.

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- **Sessions:**
  - **M:** Session ID must contain the CPO ID e.g. NLEVN1234ABC… (only Uppercase char. supported)
  - **M:** auth_id must contain a valid eMI3 Contract ID (**NLEVNC**1234ABCD)*
  - Currently Sessions are only applicable/accepted on OCPI locations and aren‘t supported for OCHP Locations
  - A session with the state of ‘INVALID’ or ‘CLOSED’ is not patchable anymore.
- **Commands:**
  - **M:** auth_id must contain a valid eMI3 Contract ID (**NLEVNC**1234ABCD)*
  - Currently only applicable/accepted on OCPI locations (not live yet)
  - Commands request will be validated against roaming and if the location is valid before being forwarded to the respective CPO

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